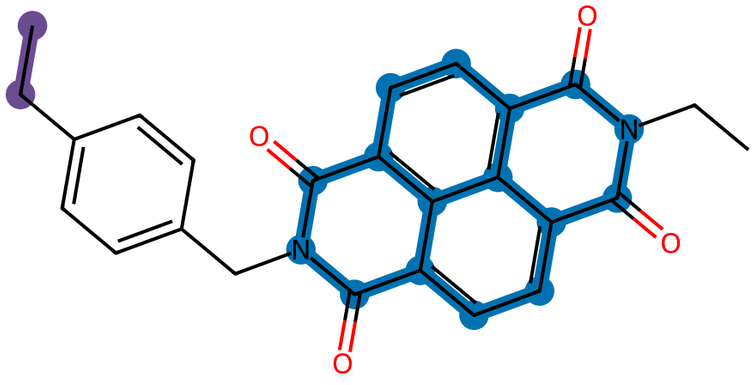
How the oligomers are modeled

Exact H-capped polystyrene fragments with pendant NDI groups; all depicted states are neutral.

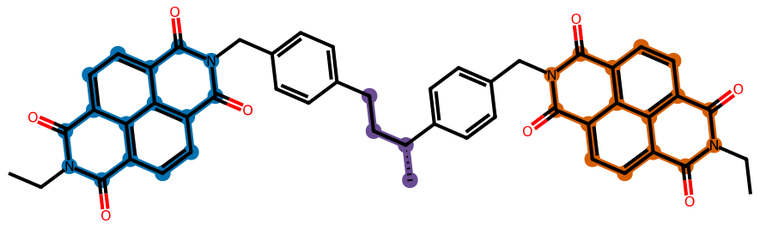
A Monomer



1 repeat unit · 1 redox site

51 atoms including H; H omitted in drawing

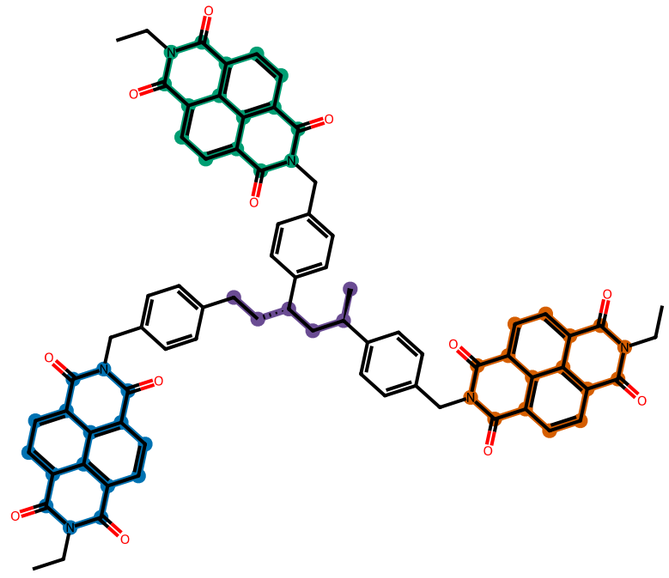
B Dimer



2 repeat units · 2 redox sites

100 atoms including H; H omitted in drawing

C Trimer



3 repeat units · 3 redox sites

149 atoms including H; H omitted in drawing

Trimer backbone detail: one open linear chain, not a triangle (schematic):

Ar_i = p-C_6H_4–CH_2–NDI. Three pendant groups; six backbone carbons, five C–C bonds, two ends.

Separate-segment model: two intact monomer segments, in parallel and antiparallel starting poses.

DFT optimization: r2SCAN-D4/SVP/SMD(MeCN) → energy: ωB97M-V/TZVPD/SMD(MeCN).

Purple: open polymer backbone. H caps are implicit. Current pilot uses 1–3 repeats; no whole-bead models.